

The Immune System

Human Physiology - Chapter 15

Teaching Materials

December 24, 2025

Outline

15.1 Defense Mechanisms

15.2 Functions of B Lymphocytes

15.3 Functions of T Lymphocytes

15.4 Active and Passive Immunity

15.5 Tumor Immunology

15.6 Diseases Caused by the Immune System

Summary

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Summary

Overview of Immune System

- The immune system provides defense against pathogens (病原体) and foreign substances.
- Two main categories of immunity:
 - **Innate immunity** (固有免疫): Nonspecific, immediate response.
 - **Adaptive immunity** (适应性免疫): Specific, develops over time with memory.
- Immune responses are coordinated by various cell types and chemical signals.

Innate Immunity Recognition

- **Pathogen-associated molecular patterns (PAMPs)**, 病原体相关分子模式) are unique molecules on pathogens.
- Examples: **lipopolysaccharide (LPS)**, 脂多糖) from gram-negative bacteria, **peptidoglycan** (肽聚糖) from gram-positive bacteria.
- **Toll-like receptors (TLRs)**, Toll 样受体) on immune cells recognize PAMPs.
- TLR activation triggers cytokine (细胞因子) secretion and immune cell recruitment.

Physical and Chemical Barriers

- First line of defense: physical and chemical barriers prevent pathogen entry.
- Skin provides physical barrier with antimicrobial secretions containing **lysozyme** (溶菌酶).
- Digestive tract uses high stomach acidity and protective bacterial population.
- Respiratory tract employs mucus secretion, ciliary movement (纤毛运动), and alveolar macrophages (肺泡巨噬细胞).



Phagocytic Cells Overview

- **Phagocytosis** (吞噬作用) is the engulfment and destruction of pathogens by specialized cells.
- Three major groups of phagocytic cells protect the body:
 - **Neutrophils** (嗜中性粒细胞): First responders from blood.
 - **Mononuclear phagocyte system** (单核吞噬细胞系统): Monocytes and macrophages.
 - **Organ-specific phagocytes** (器官特异性吞噬细胞): Specialized cells in liver, lungs, brain.

Distribution of Phagocytic Cells

- Phagocytic cells are strategically distributed throughout the body.
- **Neutrophils** (嗜中性粒细胞) circulate in blood and migrate to infection sites.
- **Kupffer cells** (库普弗细胞) in liver filter blood.
- **Alveolar macrophages** in lungs protect respiratory system.
- **Microglia** (小胶质细胞) in brain defend central nervous system.



Leukocyte Extravasation

- **Extravasation** (**diapedesis**, 渗出/穿越) is the process by which leukocytes (白细胞) exit blood vessels.
- Occurs through postcapillary venules (毛细血管后微静脉).
- Sequential steps: rolling, capture, adhesion, activation, and extravasation.
- Bacteria secrete chemicals that attract and activate white blood cells.
- Requires specific molecular interactions between leukocyte and endothelial cell surface molecules.



Phagocytosis Mechanism

- Phagocytic cells extend **pseudopods** (伪足) around particles to engulf them.
- Particle becomes enclosed in membrane-bound food vacuole.
- Vacuole fuses with **lysosomes** (溶酶体) containing digestive enzymes.
- **Lysosomal enzymes** (溶酶体酶) digest engulfed material.
- Premature lysosome fusion releases enzymes into tissue, contributing to inflammation.



Interferon Functions

- **Interferons** (干扰素) are cytokines providing nonspecific antiviral defense.
- Three types: **alpha** (**IFN- α** , α 干扰素), **beta** (**IFN- β** , β 干扰素), and **gamma** (**IFN- γ** , γ 干扰素).
- Functions include interfering with viral replication (病毒复制) and enhancing immune cell activity.
- Stimulate expression of **MHC molecules** (主要组织相容性复合体分子) on cell surfaces.



HIV Life Cycle

- **Human immunodeficiency virus (HIV, 人类免疫缺陷病毒)** targets immune cells.
- Virus binds to CD4 receptor (CD4 受体) on host cell surface.
- Uses **reverse transcriptase** (逆转录酶) to synthesize DNA from RNA genome.
- Viral DNA integrates into host cell DNA.
- Directs synthesis and assembly of new viral particles.



HIV Life Cycle

- Understanding viral life cycle is crucial for developing antiviral therapies.

Adaptive Immunity Overview

- **Adaptive immunity** (适应性免疫) provides specific defense against particular **antigens** (抗原).
- Requires prior exposure to develop effective response.
- Two types: **humoral immunity** (体液免疫) mediated by B cells and antibodies, **cell-mediated immunity** (细胞介导免疫) mediated by T cells.
- Characterized by specificity and immunological memory.

B and T Lymphocyte Comparison

- B lymphocytes (**B cells**, B 淋巴细胞) mature in bone marrow (骨髓) and produce antibodies.
- T lymphocytes (**T cells**, T 淋巴细胞) mature in thymus (胸腺) and mediate cellular immunity.
- Both have surface receptors specific for particular antigens.
- Each lymphocyte clone recognizes one specific antigen (**clonal specificity**, 克隆特异性).



Inflammatory Response Events

- **Inflammation** (炎症) is coordinated response to tissue injury or infection.
- Cardinal signs: redness (红), heat (热), swelling (肿), pain (痛).
- Chemical mediators include **histamine** (组胺), **prostaglandins** (前列腺素), and **leukotrienes** (白三烯).
- **Vasodilation** (血管舒张) and increased **capillary permeability** (毛细血管通透性) cause edema (水肿).



Local Inflammation Process

- Antigens on bacterial surface bind to antibodies, coating the bacteria.
- Antibody-antigen complexes activate **complement** (补体) proteins.
- Complement activation promotes phagocytosis by neutrophils and macrophages.
- **Histamine** from **mast cells** (肥大细胞) causes vasodilation and increased permeability.
- Results in characteristic inflammation signs: redness, heat, swelling, and pain.



Leukocyte Infiltration Timeline

- Different leukocyte types arrive at inflammation site in temporal sequence.
- **Neutrophils** (嗜中性粒细胞) arrive first within hours of injury.
- **Monocytes** (单核细胞) peak around 24-48 hours, transform into macrophages.
- **Lymphocytes** (淋巴细胞) arrive later and may persist for days.
- Sequential recruitment ensures appropriate immune response at each stage.



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Summary

B Cell Activation and Differentiation

- B lymphocytes have surface antibodies functioning as antigen receptors.
- Antigen binding to surface antibodies stimulates B cell activation.
- Activated B cells undergo clonal expansion through cell division.
- Progeny differentiate into two cell types:
 - **Plasma cells** (浆细胞): Secrete large amounts of antibodies.
 - **Memory cells** (记忆细胞): Provide long-term immunological memory.



Antibody Structure

- **Antibodies** (**immunoglobulins**, 抗体/免疫球蛋白) are Y-shaped proteins with four polypeptide chains (多肽链).
- Two **heavy (H) chains** (重链) and two **light (L) chains** (轻链).
- **Variable (V) regions** (可变区) differ between antibodies and bind specific antigens.
- **Constant (C) regions** (恒定区) are similar and determine antibody class.
- Two **Fab fragments** (Fab 片段) bind antigens; one **Fc fragment** (Fc 片段) mediates effector functions.



Antigen-Antibody Binding

- Antigen-binding site has complementary shape (互补形状) to specific antigen.
- **Fab portion** of antibody determines specificity through variable region structure.
- X-ray diffraction (X 射线衍射) reveals precise molecular interactions.
- Lock-and-key fit ensures high specificity of antibody-antigen recognition.
- This molecular complementarity is basis for immune system specificity.



Agglutination

- **Agglutination** (凝集) is the clumping together of particles by antibodies.
- Demonstrated using red blood cells in immunoassay (免疫測定).
- Antibodies bind to antigens (抗原) on cell surfaces.
- Cross-linking of multiple cells forms visible clumps.
- This technique is basis for blood typing and other diagnostic tests.



Immunoglobulin Classes

- Five classes of antibodies (**immunoglobulins**, 免疫球蛋白): IgG, IgM, IgA, IgD, IgE.
- **IgG**: Most abundant; provides long-term immunity; crosses placenta (胎盘).
- **IgM**: First antibody in primary response; pentamer structure (五聚体).
- **IgA**: Found in secretions (saliva, tears, milk); provides mucosal immunity (黏膜免疫).



Immunoglobulin Classes

- **IgE**: Binds mast cells; mediates allergic reactions and parasite defense.

Complement Fixation

- **Complement fixation** (补体固定) is activation and binding of complement proteins to pathogens.
- Antibody-antigen complex causes complement protein C4 to split into C4a and C4b.
- C4b subunit attaches to pathogen membrane.
- This triggers cascade reaction (级联反应) activating additional complement proteins.



Complement Fixation

- Complement proteins attach to membrane, marking pathogen for destruction.

Membrane Attack Complex

- **Membrane attack complex (MAC, 膜攻击复合物)** forms from complement proteins C5b through C9.
- Proteins assemble in target cell membrane creating large pore.
- Pore punctures membrane integrity.
- Results in **osmotic lysis** (渗透性裂解) as water enters cell uncontrollably.
- Direct mechanism for destroying bacterial cells and other pathogens.



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T and B Cell Interactions

- Both B and T lymphocytes respond to same antigen but play different roles.
- Antigen stimulates production of both B and T lymphocyte clones (克隆).
- **Helper T cells** (辅助性 T 细胞) enhance B cell responses.
- **Suppressor/Regulatory T cells** (抑制性/调节性 T 细胞) modulate B cell activity.
- Balanced interactions ensure appropriate immune response strength.



Dendritic Cell Migration

- **Antigen-presenting dendritic cells** (抗原呈递树突状细胞) migrate to secondary lymphoid organs (次级淋巴器官).
- Present antigens to T cells, initiating T cell activation.
- Activated T cells undergo clonal expansion by cell division.
- Some activated T cells migrate into blood circulation.
- **Chemoattractant molecules** (化学引诱分子) from inflammation guide T cells to infection site.



MHC and Coreceptors

- T cells recognize antigens only when presented with **MHC molecules** (主要组织相容性复合体分子).
- **Class-1 MHC** (MHC-I 类) found on all nucleated cells; recognized by CD8+ killer T cells.
- **Class-2 MHC** (MHC-II 类) found on antigen-presenting cells; recognized by CD4+ helper T cells.
- **Coreceptors** (共受体) ensure each T cell type interacts with appropriate MHC class.



MHC and Coreceptors

- Dual recognition provides specificity and prevents autoimmune responses.

Antigen Presentation to Helper T Cells

- **Antigen-presenting cell**
(macrophage) displays foreign antigen bound to class-2 MHC molecule.
- **Helper T cell** with CD4 coreceptor recognizes antigen-MHC complex.
- **T cell receptor (TCR, T 细胞受体)** requires this dual recognition for activation.
- Costimulatory signals between cells complete activation process.
- Activated helper T cell then regulates other immune responses.



Killer T Cell Function

- **Killer T cell** (杀伤性 T 细胞) destroys virus-infected cells.
- Recognition requires both foreign antigen and class-1 MHC molecule on infected cell surface.
- CD8 coreceptor facilitates binding to class-1 MHC.
- Dual recognition ensures killer T cells only destroy truly infected self-cells.
- Prevents inappropriate destruction of healthy cells.



Helper T Cell Activation of Killer T Cells

- Macrophage presents viral antigen with class-2 MHC to activate helper T cell.
- Activated helper T cell secretes **interleukin-2 (IL-2)** (白细胞介素-2).
- IL-2 promotes proliferation (增殖) of activated killer T cells.
- Expanded killer T cell population destroys virus-infected cells.
- Demonstrates cooperation between different immune cell types.



Helper T Cell Activation of B Cells

- Macrophage activates helper T cell by presenting antigen with class-2 MHC.
- Activated helper T cell interacts with B cell specific for same antigen.
- Helper T cell provides costimulatory signals to B cell.
- B cell divides to form clone, differentiating into plasma cells.
- Plasma cells secrete antibodies, illustrating cooperation in **humoral immunity** (体液免疫).



Cytokines Regulating Immunity

- **Cytokines** (细胞因子) are signaling molecules coordinating immune responses.
- **Interleukins** (白细胞介素) regulate lymphocyte proliferation and differentiation.
- **Interferons** modulate antiviral responses and immune cell activity.
- **Tumor necrosis factor (TNF)** (肿瘤坏死因子) promotes inflammation.
- Each cytokine has multiple effects, creating complex regulatory network.



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Virulence vs. Antigenicity

- **Virulence** (毒力) is pathogen's ability to cause disease.
- **Antigenicity** (抗原性) is pathogen's ability to stimulate immune response.
- **Active immunity** (主动免疫) can be gained through infection with virulent pathogen.
- **Vaccination** (疫苗接种) uses attenuated (减弱) pathogens with reduced virulence but preserved antigenicity.



Virulence vs. Antigenicity

- Subsequent exposure to virulent pathogen met with protective immune response.

Primary and Secondary Responses

- **Primary immune response** (初次免疫应答) occurs upon first antigen exposure with 5-10 day lag period.
- Initially produces IgM antibodies; peak levels moderate.
- **Secondary immune response** (再次免疫应答) upon re-exposure shows 1-3 day lag period.
- Produces mainly IgG antibodies; much higher peak levels that persist longer.



Primary and Secondary Responses

- Faster, stronger secondary response due to **memory cells** (记忆细胞) from primary response.

Clonal Selection Theory

- **Clonal selection theory** (克隆选择学说) explains specific antibody production.
- Each B lymphocyte expresses unique antigen receptor on surface.
- Specific antigen selects and activates only matching lymphocytes.
- Selected B cells undergo **clonal expansion** (克隆扩增) producing many identical cells.
- Most become memory cells; some become antibody-secreting plasma cells.



Clonal Selection Process

- Antigen selects lymphocytes with matching surface receptors.
- Selected cells undergo proliferation producing clone of identical cells.
- Clonal expansion occurs before differentiation into effector cells.
- Process explains how immune system generates specific responses to virtually any antigen.
- **Gene rearrangement** (基因重排) during lymphocyte development creates receptor diversity.



Active vs. Passive Immunity

- **Active immunity** (主动免疫) develops when body produces own antibodies through infection or vaccination.
- Slow onset but provides long-lasting protection with immunological memory.
- **Passive immunity** (被动免疫) involves transfer of preformed antibodies from another source.
- Provides immediate protection but temporary; no memory development.



Active vs. Passive Immunity

- Examples: maternal antibodies crossing placenta (胎盘) or gamma globulin (丙种球蛋白) injection.

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Immune Surveillance Against Cancer

- **Immunological surveillance** (免疫监视) detects and destroys cancer cells.
- Killer T cell contacts cancer cell requiring specific antigen interaction.
- T cell releases **lymphokines** (淋巴因子) including **perforin** (穿孔素) toxins.
- Perforin creates pores in cancer cell membrane causing cell death.
- Cellular immunity plays crucial role in preventing cancer development.



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Autoimmune Disease Examples

- **Autoimmune diseases** (自身免疫性疾病) occur when immune system attacks self-tissues.
- **Hashimoto's disease** (桥本氏病): Autoantibodies against thyroid.
- **Graves' disease** (格雷夫斯病): Antibodies stimulate excessive thyroid hormone production.
- **Systemic lupus erythematosus** (**SLE**, 系统性红斑狼疮): Antibodies against nuclear components.



Autoimmune Disease Examples

- **Myasthenia gravis** (重症肌无力):
Antibodies block acetylcholine
receptors (乙酰胆碱受体).

Immediate vs. Delayed Hypersensitivity

- **Immediate hypersensitivity** (速发型超敏反应) mediated by IgE antibodies, occurs within minutes.
- Involves mast cell degranulation (脱颗粒) releasing histamine and other mediators.
- **Delayed hypersensitivity** (迟发型超敏反应) mediated by T cells, occurs in 1-3 days.
- Examples: anaphylaxis (过敏性休克) vs. contact dermatitis (接触性皮炎).
- Different mechanisms require different treatment approaches.



Mechanism of Allergic Reactions

- **Allergy** (变态反应) produced when IgE antibodies attach to tissue mast cells.
- **Allergens** (变应原) are antigens that provoke allergic reactions.
- Allergen binding cross-links IgE on mast cell surface.
- Cross-linking triggers mast cell **degranulation** (脱颗粒).
- Released mediators (**histamine**, 组胺; **leukotrienes**, 白三烯) cause allergy symptoms.



Skin Test for Allergy

- Allergen injected into skin of sensitive individual.
- **Immediate hypersensitivity** produces **flare-and-wheal response** (红斑和风团反应) within minutes.
- Wheal is raised, pale area; flare is surrounding redness.
- Demonstrates IgE-mediated mast cell degranulation in skin.
- Used clinically to identify specific allergens causing patient's symptoms.



Common Environmental Allergens

- **Ragweed** (豚草) pollen causes hay fever (花粉热) in susceptible individuals.
- **House dust mites** (屋尘螨) produce waste particles triggering allergic rhinitis (变应性鼻炎) and asthma (哮喘).
- Scanning electron microscopy reveals microscopic structure of these allergens.
- Environmental control reduces allergen exposure.



Common Environmental Allergens

- Understanding common allergens helps in diagnosis and prevention strategies.

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- Immune system provides multi-layered defense: innate and adaptive immunity.
- Innate immunity offers immediate nonspecific response through physical barriers, phagocytes, and chemical mediators.
- Adaptive immunity develops specific responses with memory through B and T lymphocytes.
- B cells produce antibodies for humoral immunity; T cells mediate cellular immunity.
- Immune dysregulation causes autoimmune diseases, allergies, and immune complex diseases.
- Understanding immune mechanisms essential for developing vaccines, immunotherapies, and treating immune disorders.